

## Grade 3–6 Science

**Biology • Chemistry • Physics • Environmental Science • Psychology**

### Program Overview – [The StoryBee Foundational Series](#)

Our Grade 3–6 Science program is designed for upper elementary and early middle school students. Through richly illustrated storybooks, interactive experiments, and hands-on inquiry activities, children are introduced to the core concepts of natural sciences and psychology. The curriculum builds an interconnected framework of knowledge and scientific thinking, preparing students for deeper study in secondary school and advanced programs such as AP, IB, and A-Level.



## **Core Subjects and Learning Value**

### **Biology**

*Learning Value:* Helps children understand the origin and diversity of life, core ideas about cells, genetics, and evolution, while developing scientific inquiry skills and a respect for living things.

#### **Key Topics:**

- The Origin of Life
- Microscopic Life
- Water Properties
- Cells
- DNA & RNA
- Traits
- The Cell Cycle
- Cellular Respiration
- Food Chains
- The Energy Cycle of Nature
- The Nervous System
- The Story of the Dodo
- Evolution

## **Chemistry**

*Learning Value:* Builds understanding of the composition and transformation of matter, strengthens observational and analytical skills, and lays the groundwork for future chemistry learning.

### **Key Topics:**

- What is Chemistry
- Physical and Chemical Properties
- States of Matter
- The Atom
- Elements
- The Story of Oxygen
- The Periodic Table
- Chemical Bonding
- Chemical Reactions
- Solutions
- Acids and Bases
- Photosynthesis

## **Physics**

*Learning Value:* Encourages children to use scientific methods to explain natural phenomena, grasp key principles of forces, energy, waves, electricity, and magnetism, and apply logical reasoning to solve problems.

### **Key Topics:**

- What is Physics
- What is a Force
- Mass and Weight
- Thrust, Drag, and Gravity
- Motion
- Newton's Laws
- Friction and Air Resistance
- What is Energy
- Kinetic and Potential Energy
- Heat and Thermal Energy
- Conservation of Energy
- Work and Power
- Electricity
- Solar Power
- Wind Power
- Magnets and Magnetism
- What is a Wave
- Sound Waves
- Light Waves and Colors

## **Environmental Science**

*Learning Value:* Develops awareness of Earth's systems and ecological balance, fosters environmental responsibility, and strengthens scientific observation and data analysis skills.

### **Key Topics:**

- Ecosystems
- Climate Change
- The Atmosphere
- The Sun
- Weather
- Symbiosis
- Biodiversity
- Adaptations
- The Carbon Cycle
- The Nitrogen Cycle
- Invasive Species
- Clear-cutting
- What is Pollution

## **Psychology**

*Learning Value:* Helps children understand themselves and others, explore emotions, motivation, and social behavior, and develop empathy, communication skills, and positive relationships.

### **Key Topics:**

- History of Psychology
- Feelings and Emotions
- Empathy
- Approaches to Psychology
- The Domains of Psychology
- The Brain
- Consciousness
- How We Learn
- Memory
- Motivation
- Stress
- Cognitive Development
- Personality
- Sleep and Dreams
- Creativity

## Learning Outcomes

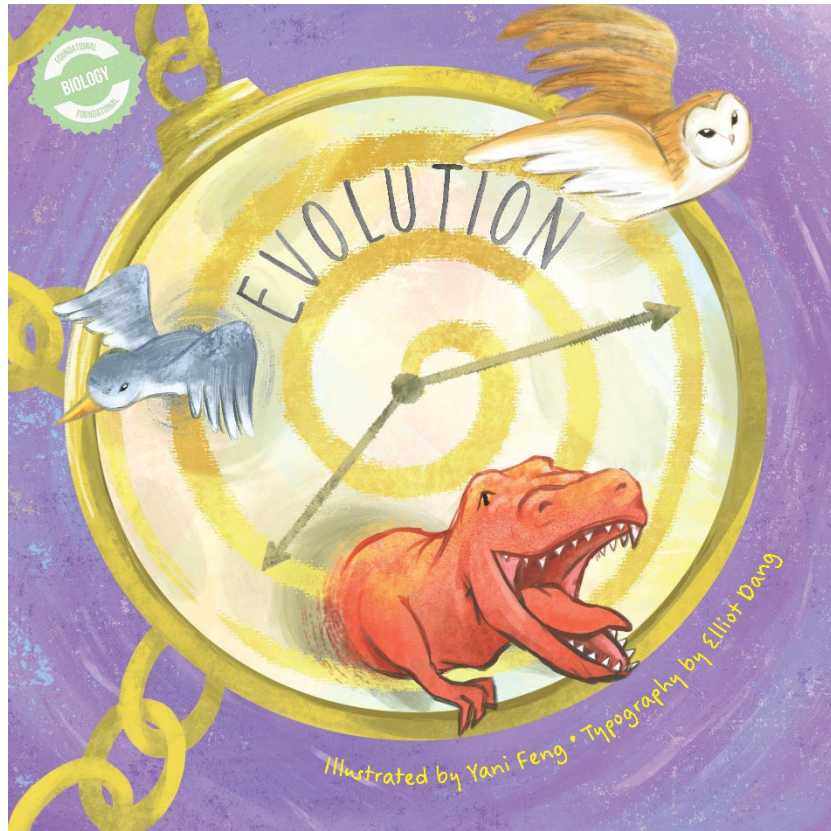
By the end of this program, students will:

- Develop a strong foundation in natural sciences and psychology, understanding the principles behind how the world works.
- Apply cross-disciplinary thinking, linking science knowledge with real-world social and environmental issues.
- Use the scientific method—observation, hypothesis, experimentation, verification—to explore and explain phenomena.
- Build environmental awareness and scientific literacy that supports sustainable living.

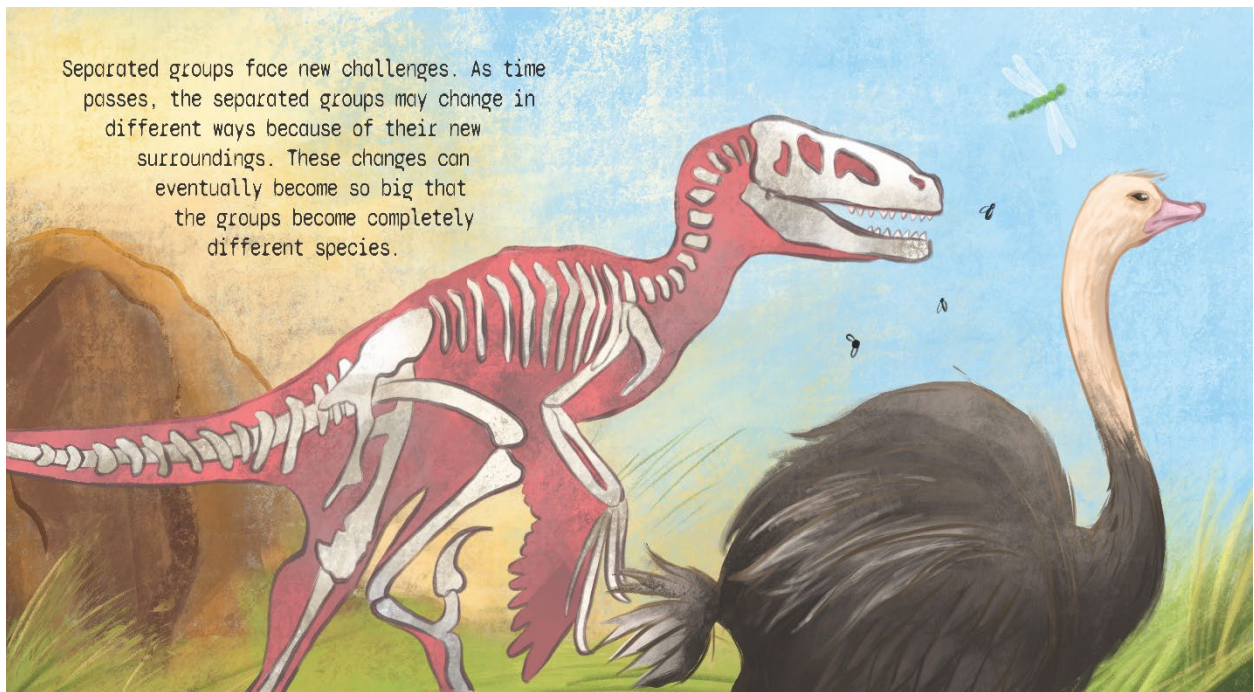
## Teaching Approach

- **One Core Theme per Lesson:** Each class begins with a storybook to make complex topics accessible.
- **Hands-On Experiments:** Students verify concepts through experiments and observation logs.
- **Cross-Disciplinary Integration:** Connects biology, chemistry, physics, environmental science, and psychology into a unified learning network.
- **Whole-Skill Development:** Incorporates math thinking, logical reasoning, science writing, and public speaking to ensure students not only understand concepts but can communicate them confidently.

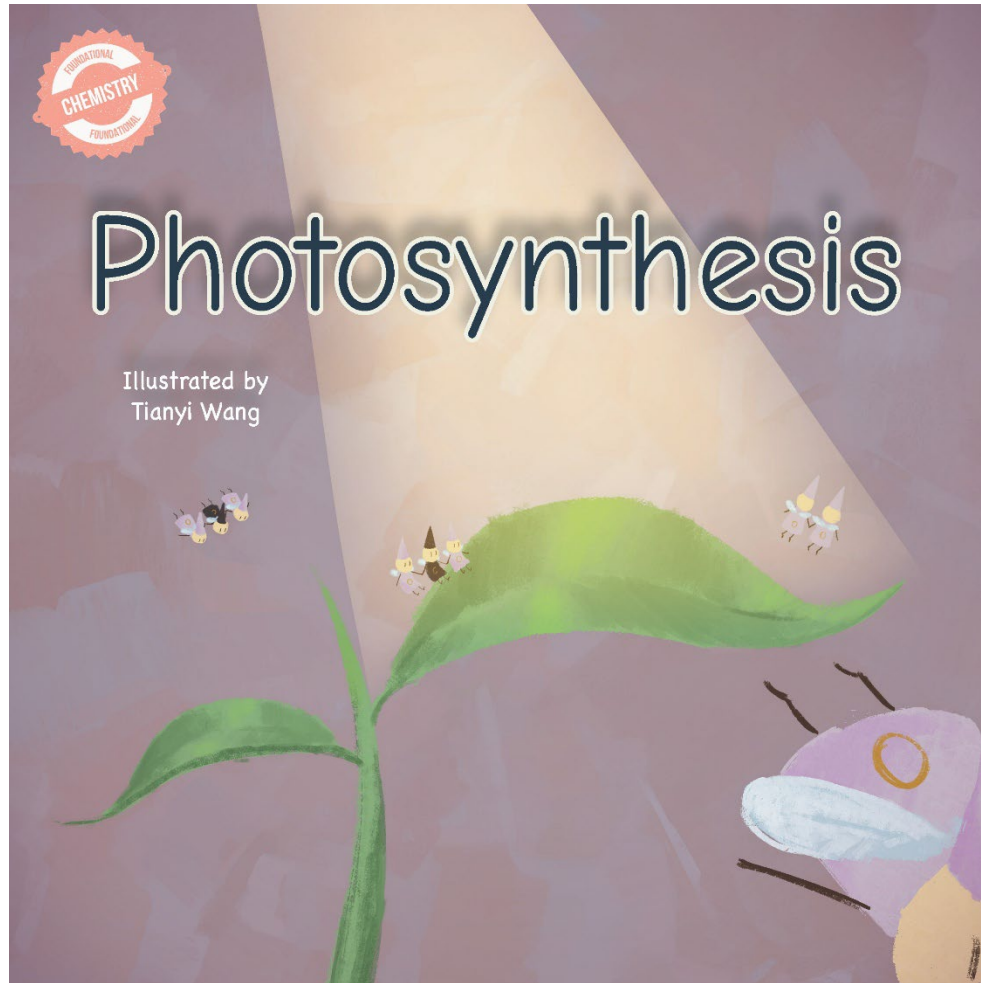
## 《Evolution》



Separated groups face new challenges. As time passes, the separated groups may change in different ways because of their new surroundings. These changes can eventually become so big that the groups become completely different species.



## 《光合作用》



A plant takes **water** from the soil, **carbon dioxide** from the air, and **sunlight** from the sky, and mixes them to create sugar (called **glucose**) that gives the plant energy to grow.

